

2.8c

Intelligence & Stability

Stability of, and influence on, intelligence

How does int. test performance change overtime?

Most children, int. tests before 3 only modestly predict future

By 4, performance begins predicting adolescent & adult scores

Consistency of scores increases over time

This was discovered testing same cohort in longitudinal studies

People more intelligent live healthier, longer

Because:

- Int. helps get more education, better jobs, healthier env.
- Int. encourages healthy living
- Prenatal events, early childhood illness influence both health & int.
- "well-wired body" factors both int. & longevity

How does age affect int.?

Cross-sectional studies shown older adults gave less correct answers on int. tests than younger adults

But longitudinal studies of same cohorts showed that int. actually remained stable, sometimes scores even increase.

Aging leads to losses & gains

G_c will $\uparrow$, as vocab + knowledge $\uparrow$

Authors, historians, philosophers produce best work later in life, after building G_c

G_f will $\downarrow$ as mental speed $\downarrow$

This is why mathematicians, scientists & chess players do best in their 20s-30s, during max G_f

But older adults' social reasoning skills $\uparrow$

Decisions get less distorted by strong negative emotions

Heritability is portion of variation that can be attributed to genes.

It's not portion of traits that can be attributed to genes.

Estimates of heritability of intelligence range 50% 80%

Heritability varies in changing envs.

in similar env. heritability $\uparrow$, becoming main source of differences

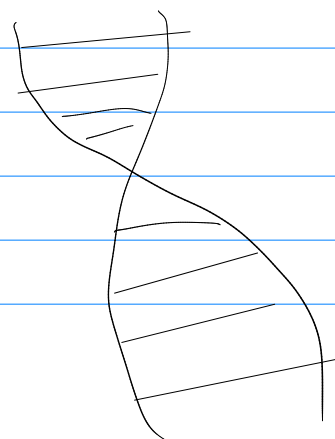
in different env., heritability $\downarrow$

Identical twins share intelligence, & specific talents (twin studies)

But there is no known 'genius' gene

gene variations accounted for $\sim 5\%$ of differences in educational achievement

intelligence is poly genetic



Env. & Int.

Dz twins: no more genetically alike than other siblings, but share env. & often treated similarly.

Their int. tests are more alike than those of nontwins

Showing env. has some effect too.

Adoption from poverty into financially secure (obviously) $\rightarrow$ int. $\uparrow$

Adoption of mistreated/neglected children $\rightarrow$ int. score $\uparrow$

Int. scores of virtual twins - same age, unrelated adopted as infants, raised as siblings, correlate positively: $+0.28$

Adopted kid's int. score resembles bio family more than adopted family

Over time, adopted kid's verbal ability score even more like bio parents

In MZ twins, heritability of g $\uparrow$ /time.

middle school: 41%

adolescence: 55%

young adulthood: 66%

How do genes + env. interact?

Genes shape the experiences that shape us.

ex. natural aptitude for school $\rightarrow$ P(stay in school, read, etc.) $\uparrow$
brain power $\uparrow \quad \Leftarrow$

Small genetic advantage $\rightarrow$ env. changes that multiply advantage

Env. can work in reverse, depressing cognitive development

Studies show, children neglected in orphanages had extreme deprivation of care, crushing natural int.

This shows, poverty can depress cognitive development & produce stresses impeding cognitive performance.

However, no env. that fast-forwards a normal infant into genius

Nutrition $\uparrow \rightarrow$ int. score $\uparrow$

Growth Mind set

Growth mindset: believing int. can change

Fixed mindset: believing int. is fixed

motivation can affect int. test performance.

growth mindset makes u more likely to do well

Important not to overstate power of growth:
children may blame selves for struggle.

Mindset isn't everything, but it matters

Personal Connection.

One trait of mine I care most about is my "intelligence". There are definitely genetic components to my performance in school, as I've noticed my dad has similar skills to me, and I've heard I've had these skills in childhood. However, there is also a very important env. factor.

I was fortunate enough to be diagnosed with ADHD, given access to meds, and help from numerous psychological professionals. If I wasn't, my performance in school would be extremely bad, as my challenges with executive functioning & focus would, and in the past have, severely impact my performance

MCQ

1. b 2. c 3. c 4. c 5. d